



WIFIINTRUSION DETECTIONSYSTEM



A DESIGN PROJECT REPORT

Submitted by

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The work embodied in the present project report entitled —**WIFI INTRUSION DETECTION SYSTEM** has been carried out by the students ,**RAJEV SHREE U, SHAJATHI BEE M, SHIBANI S, SOWMIYAA VS**, The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

A Wi-Fi Intrusion Detection System (WIDS) is a crucial security mechanism designed to monitor, detect, and prevent unauthorized access and malicious activities within wireless networks. With the rapid growth of wireless communication technologies, networks have become increasingly vulnerable to various threats such as unauthorized access points. The proposed system utilizes advanced detection techniques, including signature-based detection and anomaly-based detection, to enhance the accuracy and effectiveness of threat identification. It captures wireless packets, examines traffic behavior, and compares it against predefined rules or normal usage profiles. When an intrusion is detected, the system generates alerts and logs the activity for further analysis and response. Additionally, the system can be integrated with existing network infrastructure to provide a scalable and flexible security solution. This Wi-Fi Intrusion Detection System aims to improve network integrity, confidentiality, and availability by proactively identifying vulnerabilities and preventing potential attacks. The implementation of such a system is essential for organizations and individuals to safeguard sensitive data and maintain secure wireless communication environment.

Keywords: Wi-Fi Security, Intrusion Detection System (IDS), Wireless Network Protection, Anomaly Detection, Signature-Based Detection, Network Traffic

Analysis, Cyber Threat Detection, Rogue Access Point Detection, Denial-of-Service (DoS).

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LIST OF ABBREVIATIONS

TCP	- Transmission Control Protocol
UDP	- User Datagram Protocol
SSID	- Service Set Identifier
WPA	- Wi-Fi Protected Access
WPA2	- Wi-Fi Protected Access 2
EAP	- Extensible Authentication Protocol
MITM	- Man-in-the-Middle
NIC	- Network Interface Card
ML	- Machine Learning
RF	- Radio Frequency
BSSID	- Basic Service Set Identifier
HTTP	- Hyper Text Transfer Protocol
HTTPS	- Hyper Text Transfer Protocol Secure

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

A Wi-Fi Intrusion Detection System (WIDS) is an advanced security solution developed to safeguard wireless networks from unauthorized access, misuse, and cyber threats. With the rapid growth of wireless communication technologies, Wi-Fi networks have become an integral part of daily life in homes, educational institutions, and organizations. However, this widespread usage also exposes networks to various vulnerabilities and attacks. The primary purpose of a WIDS is to continuously monitor wireless traffic, analyze data

transmissions, and detect any abnormal or suspicious activities that may indicate a security breach.

The system operates by capturing and inspecting wireless packets transmitted across the network using specialized monitoring tools and sensors. It examines different parameters such as packet size, transmission frequency, source and destination addresses, and protocol behavior. By applying signature-based detection methods, the system compares network activity against a database of known attack patterns, enabling it to quickly identify recognized threats. In addition, anomaly-based detection techniques are used to establish a baseline of normal network behavior and detect deviations that may indicate unknown or emerging attacks.

It also supports compliance with security standards and policies, which is essential for organizations handling sensitive information. Overall, a Wi-Fi Intrusion Detection System enhances the confidentiality, integrity, and availability of wireless networks by providing proactive and continuous security monitoring.

1.2 PROBLEM IDENTIFICATION

The Wi-Fi networks have become widely used in various environments such as homes, organizations, educational institutions, and public spaces. However, this widespread usage has also introduced significant security challenges. Unlike wired networks, wireless networks operate over an open medium, making them highly vulnerable to unauthorized access and cyber-attacks. Attackers can easily intercept signals, exploit weak security configurations, and gain access to sensitive information without being physically connected to the network.

Traditional security mechanisms such as encryption and authentication protocols are not sufficient to detect all types of attacks, especially those that are newly developed or unknown. As a result, malicious activities such as rogue access point deployment, denial-of-service attacks,

packet sniffing, and man-in-the-middle attacks often go unnoticed until significant damage has already occurred.

Public Wi-Fi environments such as airports, railway stations, hotels, and shopping malls also benefit significantly from WIDS implementation. These networks are highly prone to attacks due to open access, and the system helps in monitoring user activities and preventing malicious actions. Additionally, WIDS is used in smart homes and IoT-based environments, where multiple connected devices operate over wireless networks, increasing the risk of security vulnerabilities.

In summary, the key problem lies in the inability of existing wireless security mechanisms to provide comprehensive, real-time protection against evolving cyber threats, thereby creating the need for a more advanced and intelligent intrusion detection system.

1.3 WIRELESS EVOLUTION DOMAIN

Wireless communication has rapidly evolved over the years, transforming from basic radio-based systems into highly advanced Wi-Fi networks that support fast and reliable data transmission. Today, wireless technology plays a vital role in connecting devices such as smartphones, laptops, and smart home systems, enabling seamless communication and access to digital services. The growth of technologies like cloud computing and the Internet of Things (IoT) has further increased the dependence on wireless networks. However, as wireless systems expand and become more complex, they also introduce new security concerns due to their open and easily accessible nature, making them more vulnerable to external threats and unauthorized access. Intercepted, making them highly susceptible to unauthorized access and malicious activities.

The traditional security approaches, including passwords and encryption protocols, are no longer adequate to defend against sophisticated and evolving attacks. Cyber attackers continuously develop new techniques to bypass these defenses, such as creating fake access points, injecting malicious packets, or disrupting network services. of medical procedures. Moreover, the integration of deep learning models with medical imaging systems can enable real-time detection during endoscopic examinations.

Another major factor emphasizing the need for this system is the increasing use of wireless networks in critical sectors such as banking, healthcare, education, and government organizations. These sectors handle highly sensitive information that must be protected from unauthorized access. Any breach in security can result in loss of confidentiality, financial damage, and legal implications. provide an effective solution for improving the detection of gastrointestinal lesions.

1.3.1 NETWORK THREATS DOMAIN

The proposed approach for the Wi-Fi Intrusion Detection System (WIDS) focuses on developing a robust and intelligent mechanism to monitor, analyze, and secure wireless network environments. The system is designed to operate in real time by continuously capturing wireless traffic through network sensors or monitoring tools placed within the coverage area.

The system also incorporates a data preprocessing stage, where captured packets are filtered and organized to extract relevant features such as IP addresses, MAC addresses, packet size, and transmission frequency. These features are then analyzed using rule-based algorithms or lightweight machine learning models to detect suspicious activities. Upon identifying any anomaly or intrusion attempt, the system generates real-time alerts.

Additionally, the proposed system is designed to be scalable and

adaptable, allowing integration with existing network infrastructure such as firewalls and security management systems. A user-friendly interface is provided for network administrators to monitor network status, review alerts, and take necessary actions.

Anomaly-based detection, on the other hand, builds a dynamic baseline profile of normal network behavior by observing parameters such as bandwidth usage, connection patterns, and protocol activity over time. Any significant deviation from this baseline is flagged as suspicious. Additionally, behavior-based analysis focuses on identifying unusual user or device activities, such as repeated failed authentication attempts or abnormal access timings, which may indicate potential intrusions.

These features are then fed into the detection engine, where rule-based algorithms and machine learning techniques such as classification or clustering are applied to enhance detection accuracy and adaptability.

1.3.2 SYSTEM MONITORING DOMAIN

The Wi-Fi Intrusion Detection System (WIDS) has a wide range of applications in environments where wireless network security is a critical requirement. One of the primary applications of the system is in corporate and enterprise networks, where large volumes of sensitive data are transmitted over Wi-Fi. In such environments, WIDS helps in detecting unauthorized users, preventing data breaches, and ensuring secure communication between devices, thereby protecting valuable organizational information.

In educational institutions such as schools, colleges, and universities, the system plays a significant role in maintaining secure and controlled network access. Since a large number of students and staff connect to the network

simultaneously, the chances of misuse and cyber-attacks increase. WIDS helps administrators monitor network activity, detect suspicious behavior, and prevent unauthorized access, ensuring a safe digital learning environment.

1.3.3 REAL-TIME DETECTION DOMAIN

Real-time detection plays a crucial role in protecting wireless networks from potential cyber-attacks. Since data transmission in Wi-Fi networks occurs continuously, any delay in identifying a threat can lead to significant damage, including data loss or system disruption. A real-time intrusion detection system ensures that suspicious activities are identified instantly and appropriate actions are taken without delay. This improves the overall reliability, security, and efficiency of the network, making it capable of handling modern security challenges effectively.

In addition, real-time detection plays an important role in maintaining compliance with security standards and policies. Organizations that handle sensitive data require immediate threat identification and response to meet regulatory requirements. By ensuring constant surveillance and quick mitigation of risks, the system helps maintain the confidentiality, integrity, and availability of network resources.

1.4 AIM AND OBJECTIVE

1.4.1 Aim

The primary aim of this project is to design and develop an efficient and reliable Wi-Fi Intrusion Detection System (WIDS) that can monitor wireless network traffic, detect unauthorized access, and identify potential security threats in real time. The system is intended to enhance the overall security of wireless networks by providing continuous surveillance and intelligent analysis of network activities. It focuses on protecting sensitive data, maintaining network integrity, and ensuring uninterrupted communication by identifying and mitigating various types of cyber-attacks. The aim also includes developing a scalable and adaptable solution that can be implemented in different

environments such as organizations, educational institutions, healthcare systems, and public Wi-Fi networks.

1.4.2 Objective

The objectives of the system are to continuously capture and analyze wireless network data packets in order to identify suspicious patterns and abnormal behavior. It aims to implement both signature-based and anomaly-based detection techniques to effectively detect known as well as unknown threats. Another important objective is to generate real-time alerts and maintain detailed logs of detected intrusions, enabling quick response and further analysis by network administrators. The system also focuses on reducing false positives and improving detection accuracy through efficient data processing and intelligent algorithms. Additionally, it aims to provide a user-friendly interface for monitoring network activity and managing security events. The project further seeks to ensure compatibility and integration with existing network security infrastructure, thereby enhancing its practicality and usability. Overall, the objective is to create a proactive security mechanism that strengthens wireless network protection and ensures safe and reliable communication.

1.5 SCOPE OF THE PROJECT

The scope of this project focuses on the design and development of a comprehensive Wi-Fi Intrusion Detection System (WIDS) capable of monitoring, analyzing, and securing wireless network environments against a wide range of cyber threats. The system is intended to operate within wireless local area networks by continuously capturing and examining network traffic in real time. It covers the detection of various types of intrusions such as unauthorized access, rogue access points, denial-of-service attacks, packet sniffing, and man-in-the-middle attacks. By providing real-time monitoring and

alert mechanisms, the system ensures that potential threats are identified at an early stage and appropriate actions can be taken to prevent damage.

The project scope includes the implementation of advanced detection techniques such as signature-based detection for identifying known threats and anomaly-based detection for recognizing unusual patterns and unknown attacks. It also involves the use of data preprocessing and feature extraction methods to efficiently handle large volumes of network traffic and improve the accuracy of intrusion detection. The system is designed to maintain logs of all detected events, which can be used for analysis, auditing, and improving future security measures. This enhances the overall effectiveness of the system in managing and responding to security incidents.

Furthermore, the scope extends to developing a user-friendly interface that allows network administrators to monitor network activity, view alerts, and manage security configurations with ease. The system is also designed to be scalable and adaptable, making it suitable for deployment in various environments such as enterprises, educational institutions, healthcare organizations, and public Wi-Fi networks. It supports integration with existing security infrastructure, including firewalls and intrusion prevention systems, thereby enhancing the overall network defense mechanism.

Overall, the scope of this project encompasses the development of a reliable, scalable, and efficient Wi-Fi Intrusion Detection System that enhances wireless network security through continuous monitoring, intelligent analysis, and timely threat detection, thereby ensuring safe and secure communication in modern networking environments.

CHAPTER 2

LITERATURE SURVEY

2.1 MACHINE LEARNING-BASED INTRUSION DETECTION IN WIRELESS NETWORKS

The work by M. Tavallae et al. (2009) presents a significant contribution to intrusion detection systems through the introduction of the NSL-KDD dataset, which is widely used for evaluating machine learning-based IDS models. Intrusion Detection Systems (IDS) have evolved significantly with the integration of machine learning techniques, especially in wireless network environments. This study focuses on the application of supervised and unsupervised learning algorithms to detect malicious activities in Wi-Fi networks. Algorithms such as Decision Trees, Random Forest, Support Vector Machines (SVM), and Neural Networks are trained using benchmark datasets like KDD Cup 99 and NSL-KDD. These models learn patterns of normal and abnormal network behavior and classify incoming traffic accordingly. The research highlights that machine learning models can significantly improve detection accuracy and adaptability compared to traditional methods. One of the major advantages of this approach is its ability to detect unknown or zero-day attacks by identifying unusual patterns in network traffic.

However, the study also identifies several challenges. Machine learning models require large volumes of labeled data for training, which may not always be available. Additionally, the computational complexity of training and maintaining these models can be high, making them less suitable for lightweight or real-time applications without optimization. Another concern is overfitting, where the model performs well on training data but poorly on real-world data. Despite these limitations, the study concludes that machine learning-based IDS provides a promising direction for improving wireless network security. This approach is particularly relevant to modern Wi-Fi Intrusion Detection Systems.

2.2 SIGNATURE-BASED INTRUSION DETECTION SYSTEMS FOR WI-FI SECURITY

The concept of intrusion detection systems was first formally introduced by Dorothy E. Denning in 1987, where she proposed a model based on monitoring system activities and detecting abnormal patterns using predefined rules. This study laid the foundation for signature-based intrusion detection systems, which rely on a database of known attack patterns. However, the study also acknowledges that such systems are limited in detecting new or unknown attacks. Since the detection depends entirely on predefined signatures, any variation or new attack pattern may go unnoticed. The system compares incoming network traffic with stored signatures to identify potential threats such as denial-of-service attacks, spoofing, and unauthorized access.

Denning's model emphasizes accuracy and reliability in detecting known threats, making it highly effective in controlled environments. However, the study also acknowledges that such systems are limited in detecting new or unknown attacks. Since the detection depends entirely on predefined signatures, any variation or new attack pattern may go unnoticed. Additionally, maintaining and updating the signature database requires continuous effort. Despite these limitations, the research remains a cornerstone in the field of network security, and signature-based detection continues to be widely used in modern systems. This approach is relevant to the proposed project, where known Wi-Fi networks are used as a reference to classify safe connections. Implemented visualization techniques to highlight the regions in the image responsible for the classification. The study also emphasized that integrating such AI-based detection systems into clinical workflows can enhance the overall accuracy and reliability of gastrointestinal disease diagnosis, ultimately improving patient care and treatment outcomes.

2.3 ANOMALY-BASED INTRUSION DETECTION

The research conducted by Debarati Sen (2010) explores anomaly-based intrusion detection techniques in wireless networks. The study focuses on identifying deviations from normal network behavior by establishing a baseline of standard activity. Using statistical models and behavioral analysis, the system monitors parameters such as traffic volume, connection patterns, and signal variations. Any significant deviation from the baseline is flagged as a potential intrusion.

The study highlights the advantage of anomaly-based systems in detecting unknown and zero-day attacks, which are not present in signature databases. However, it also discusses the challenge of high false positive rates, where legitimate activities may be incorrectly classified as malicious. The research suggests that improving baseline accuracy and incorporating adaptive learning techniques can reduce false alarms. Despite its limitations, anomaly detection is considered essential for modern wireless security systems, as it provides the ability to detect evolving threats. This concept is directly applicable to the proposed system, where unusual Wi-Fi patterns such as unknown networks are flagged as medium-risk threats.

The researchers emphasized the importance of integrating deep learning models with explainable artificial intelligence techniques to improve the transparency of the Wi-Fi Intrusion process. The research suggests that improving baseline accuracy and incorporating adaptive learning techniques can reduce false alarms. Despite its limitations, anomaly detection is considered essential for modern wireless security systems, as it provides the ability to detect evolving threats. Visualization methods such as heat maps and activation maps were used to highlight the regions of the network traffic that contributed to the model's prediction. These visual explanations help clinicians understand how the model identifies Wi-Fi packets and verify the accuracy of the Intrusion.

2.4 DETECTION OF ROGUE ACCESS POINTS

The study by Arunesh Mishra et al. (2007) focuses on the detection of rogue access points in wireless networks. Rogue APs are unauthorized devices that mimic legitimate networks to deceive users and capture sensitive information. The research proposes techniques for identifying such threats by analyzing network parameters such as SSID, BSSID, and signal strength. One of the key contributions of the study is the detection of duplicate SSIDs with different MAC addresses, which is a strong indicator of a fake access point.

The authors also explore methods such as MAC address verification and signal strength comparison to improve detection accuracy. The study highlights that attackers often use similar network names to trick users, making it essential to analyze multiple parameters rather than relying on SSID alone. However, the research acknowledges that sophisticated attackers may spoof MAC addresses, making detection more challenging. Despite this, the study concludes that combining multiple detection techniques significantly enhances security. This work is highly relevant to the proposed project, where duplicate SSIDs are classified as high-risk networks.

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2.5 REAL-TIME WIRELESS NETWORK MONITORING

The research by Rashmi Shankar (2015) emphasizes the importance of real-time monitoring in wireless network security. The study discusses systems that continuously scan available Wi-Fi networks and analyze their properties to detect potential threats. Using tools such as packet analyzers and wireless scanners, the system collects data on SSID, BSSID, signal strength, and traffic patterns. This data is processed in real time to identify suspicious activities and generate alerts.

The study highlights the role of visualization tools such as dashboards, which provide users with an intuitive interface for monitoring network activity. Real-time alerts enable quick response to threats, reducing the risk of data breaches. However, the research also identifies challenges such as high resource consumption and system performance issues. Continuous monitoring requires efficient algorithms to ensure smooth operation without affecting system performance. Despite these challenges, the study concludes that real-time monitoring is essential for effective intrusion detection. This concept is reflected in the proposed system's dynamic dashboard, which updates network information automatically.

Despite these limitations, the research concludes that real-time monitoring is a critical requirement for modern wireless security systems. It enables proactive detection and response to threats, making it an essential component of intrusion detection systems. This concept is clearly reflected in the proposed system, where the dashboard updates network information dynamically and provides real-time risk classification. Additionally, ensuring the accuracy of detection while minimizing false positives is a complex task. The

study suggests the use of optimized algorithms and efficient data processing techniques to address these challenges.

2.6 HYBRID INTRUSION DETECTION SYSTEM

The study by the SANS Institute (2014) explores hybrid intrusion detection systems, which combine multiple detection techniques to improve overall performance and reliability. The research integrates signature-based and anomaly-based detection methods to create a comprehensive security solution. In this approach, known attacks are detected using signature matching, while unknown threats are identified through anomaly detection.

The study demonstrates that hybrid systems offer several advantages over standalone approaches. By combining the strengths of different techniques, hybrid systems can achieve higher detection accuracy and lower false positive rates. The research also highlights the flexibility of hybrid systems, as they can adapt to different network environments and evolving threats. However, the study also discusses challenges associated with hybrid systems, including increased complexity and higher implementation costs. Designing and maintaining such systems require careful planning and expertise. Additionally, integrating multiple detection techniques may lead to performance overhead if not optimized properly.

Despite these challenges, the research concludes that hybrid intrusion detection systems represent the future of wireless network security. They provide a balanced approach that addresses the limitations of individual techniques and offer comprehensive protection against a wide range of threats. However, the study also discusses challenges associated with hybrid systems, including increased complexity and higher implementation costs. Designing and maintaining such systems require careful planning and expertise. This concept aligns closely with the proposed Wi-Fi intrusion detection system, which combines elements of signature-based detection (known networks) and anomaly detection (unknown and duplicate networks) to classify risks effectively.

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING SYSTEM

The existing systems for Wi-Fi intrusion detection and wireless network security are primarily based on traditional security mechanisms such as encryption protocols, firewalls, and basic intrusion detection techniques. Most wireless networks rely on security standards like WPA2 (Wi-Fi Protected Access 2) and WPA3, which focus mainly on protecting data confidentiality and authentication. While these protocols ensure that data transmitted between devices is encrypted and secure from eavesdropping, they do not actively monitor or detect malicious activities occurring within the network environment. As a result, these systems are limited in their ability to identify real-time threats such as rogue access points, spoofing attacks, and unauthorized network access.

In many existing implementations, intrusion detection is performed using signature-based systems. These systems operate by maintaining a database of known attack patterns or signatures and comparing incoming network traffic against this database. When a match is found, the system generates an alert indicating a possible intrusion. Signature-based systems are effective in detecting known threats and provide high accuracy with low false positives. However, they suffer from a major limitation: they cannot detect new or unknown attacks that do not match existing signatures. In rapidly evolving wireless environments, where attackers continuously develop new techniques, this limitation becomes a significant drawback. Additionally, maintaining and

updating the signature database requires continuous effort and expert intervention. Another approach used in existing systems is anomaly-based detection. In this method, the system first establishes a baseline of normal network behavior and then identifies deviations from this baseline as potential threats. While anomaly-based systems are capable of detecting unknown attacks, they often generate a high number of false positives. This is because not all deviations from normal behavior indicate malicious activity. For example, a sudden increase in network traffic due to legitimate usage may be incorrectly flagged as an intrusion. This reduces the reliability of the system and can lead to unnecessary alerts, making it difficult for users or administrators to distinguish between real threats and false alarms.

Furthermore, many existing systems lack real-time monitoring capabilities. Network administrators often rely on manual tools such as command-line utilities or offline analysis software to monitor network activity. These tools require technical expertise and are not user-friendly for general users. The absence of real-time visualization and alert mechanisms delays the detection and response to potential threats. Traditional intrusion detection systems may not be able to handle large volumes of data efficiently, leading to performance degradation. Moreover, continuous monitoring and analysis require significant computational resources, which may not be feasible for all system



Figure 3.1 Existing sytem

3.2 DISADVANTAGES

- High cost of deployment and maintenance requires specialized hardware, software, and skilled personnel
- False positives may incorrectly flag legitimate network activity as threats
- Limited detection capability cannot always identify sophisticated or encrypted attacks
- No active prevention only detects intrusions but does not stop attacks
- Complex configuration requires careful tuning and continuous updates
- Scalability issues make it difficult to manage in large or dynamic networks
- Performance overhead consumes system resources during monitoring
- Dependence on sensor placement can create blind spots if not properly positioned

CHAPTER 4

PROBLEMS IDENTIFIED

In today's digital world, wireless networks (Wi-Fi) have become an essential part of everyday life. From educational institutions and workplaces to homes and public spaces, Wi-Fi connectivity is widely used for communication, data transfer, and internet access. However, this rapid growth in wireless usage has also introduced serious security concerns. Many users connect to available Wi-Fi networks without verifying their authenticity, making them vulnerable to various cyber threats.

One of the major problems in wireless network security is the presence of unauthorized or malicious access points. Attackers can easily create fake Wi-Fi networks, often referred to as —Evil Twinll networks, which mimic legitimate network names (SSIDs). These fake networks trick users into connecting, allowing attackers to intercept sensitive data such as passwords, personal information, and browsing activity. Since these malicious networks often appear identical to genuine ones, it becomes extremely difficult for ordinary users to differentiate between safe and unsafe connections.

Another significant issue is the lack of awareness among users regarding Wi-Fi security risks. Most users prioritize convenience over security and tend to connect to any available network without proper verification. Public places like cafes, colleges, and airports are especially vulnerable environments where multiple networks with similar names exist. This increases the chances of users connecting to unknown or harmful networks unknowingly.

Additionally, there is a lack of real-time monitoring tools available to detect suspicious Wi-Fi activities at the user level. Existing security solutions are often complex, expensive, or designed for large organizations rather than individual users or small-scale

environments. As a result, many users do not have access to simple and effective tools that can continuously monitor nearby networks and alert them about potential threats.

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8 Another problem is the difficulty in identifying duplicate network names (SSIDs) that may indicate a possible attack. When multiple networks with the same name appear, it could be a sign of a spoofing attempt. However, most devices do not provide clear warnings in such cases, leaving users unaware of the potential danger. This lack of visibility increases the risk of data breaches and unauthorized access.

Furthermore, traditional network security mechanisms mainly focus on protecting devices after they are connected to a network, rather than preventing users from connecting to unsafe networks in the first place. This reactive approach is not sufficient in scenarios where the threat originates from the network itself.

There is also a challenge in maintaining a record of suspicious activities. Without proper logging and tracking, it becomes difficult to analyze patterns, identify repeated threats, or take preventive measures. Users and administrators need a system that not only detects risks but also stores alert information for future reference.

Considering all these issues, there is a clear need for a simple, real-time, and user- friendly system that can monitor nearby Wi-Fi networks, identify unknown and potentially harmful connections, and alert users effectively. Such a system should help users make informed decisions before connecting to any network, thereby reducing the risk of cyber attacks.

The problem addressed in this project is the absence of an efficient and accessible solution for detecting Wi-Fi intrusions and suspicious networks in real time. The goal is to bridge this gap by providing a system that continuously scans available networks, classifies their risk levels, and presents the information through an intuitive dashboard. This approach ensures that users are not only aware of potential threats but are also equipped with the necessary information to stay secure in wireless environments.

CHAPTER 5

PROPOSED SYSTEM

5.1 PROPOSED SYSTEM

The proposed system aims to develop an automated and intelligent solution for detecting gastrointestinal lesions using deep learning techniques. Gastrointestinal diseases such as ulcers, polyps, bleeding, and tumors require early detection to prevent serious health complications. Traditional diagnostic methods rely heavily on manual analysis of endoscopic images by experienced clinicians. Although this approach is effective, it can be time-consuming and prone to human error due to fatigue or variability in interpretation. The proposed system utilizes deep learning models, particularly Convolutional Neural Networks (CNNs), to analyse endoscopic images and identify patterns associated with gastrointestinal lesions. Deep learning models are capable of learning complex visual features from large datasets, making them highly suitable for medical image analysis.

The system is designed to assist clinicians by automatically analysing images and highlighting suspicious regions that may indicate the presence of lesions. By doing so, the system can improve diagnostic accuracy and reduce the workload for healthcare professionals. The first stage of the proposed system involves collecting a large dataset of endoscopic images. These images are obtained from publicly available medical datasets such as Kvasir and HyperKvasir, which contain high-quality images representing various gastrointestinal conditions. The dataset includes images of normal mucosa as well as abnormal conditions such as polyps, ulcers, bleeding regions, and inflammatory tissues.

The collected dataset serves as the foundation for training the deep learning model. Each image in the dataset is labelled according to the type of

gastrointestinal condition it represents. Proper labelling of images is essential for supervised learning, as it enables the model to learn the differences between normal and abnormal tissues. A diverse dataset is important for improving the generalization capability of the model. Images captured under different lighting conditions, camera angles, and anatomical variations help the model learn a wide range of visual patterns. This ensures that the system can perform accurately even when analysing new unseen images.

To address these issues, several preprocessing techniques are applied. First, the images are resized to a standard resolution so that they can be efficiently processed by the deep learning model. Resizing ensures uniformity in image dimensions and reduces computational complexity during training. Next, pixel values are normalized to maintain consistent intensity levels across the dataset. Normalization helps the model learn more effectively by preventing large variations in pixel values.

5.2 PROPOSED SYSTEM ARCHITECTURE

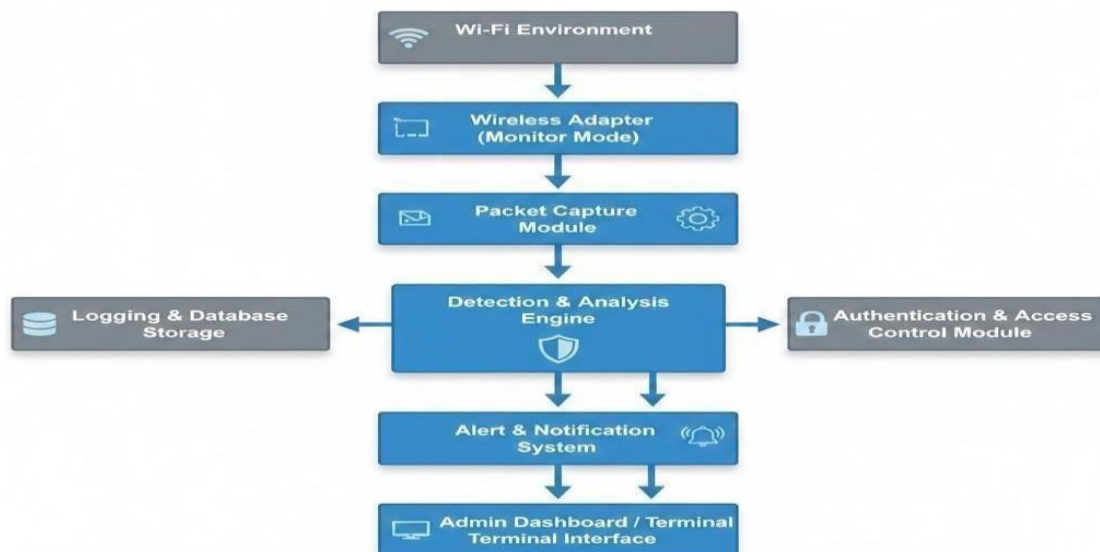


Figure.5.2 Proposed System

5.3 ADVANTAGES

- Detects unauthorized access points and rogue devices in the wireless network
- Identifies suspicious activities such as unusual traffic patterns or attacks
- Enhances overall network security by continuously monitoring wireless communication
- Provides real-time alerts to administrators for quick response
- Helps in enforcing security policies across the network
- Improves visibility of all devices connected to the Wi-Fi network
- Assists in detecting common wireless attacks like spoofing and sniffing
- Supports forensic analysis by maintaining logs of network activity
- Can be integrated with existing security systems for better protection
- Reduces the risk of data breaches and unauthorized access

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

The proposed Wi-Fi Intrusion Detection System is designed to be lightweight and does not require any specialized or high-end hardware components. It can be implemented using commonly available computing devices with basic networking capabilities.

A standard computer or laptop is required to run the application. The system should have at least a dual-core processor (Intel i3 or equivalent) to ensure smooth execution of the scanning and monitoring processes. Higher performance processors can improve the speed of data processing and real-time updates.

A minimum of 4 GB RAM is recommended for efficient performance. This ensures that the system can handle background processes such as Wi-Fi scanning, running the web server, and updating the dashboard without lag. For better performance, especially during continuous monitoring, 8 GB RAM is preferred.

The system requires minimal storage space as it primarily stores text-based logs and program files. A minimum of 500 MB of free disk space is sufficient. However, additional storage may be required if the alert logs grow over time.

A wireless network adapter is essential for scanning nearby Wi-Fi networks. The system should have a built-in or external Wi-Fi adapter that supports network scanning. For advanced monitoring and better accuracy, a Wi-Fi adapter capable of monitor mode is recommended, although not mandatory for basic functionality.

While the system primarily scans local Wi-Fi networks, an internet connection is required for certain features such as sending OTP emails for user authentication. Without internet access, the core scanning functionality will still work, but email-based alerts may not function.

6.2 SOFTWARE REQUIREMENTS

The proposed Wi-Fi Intrusion Detection System is developed using a combination of programming languages, frameworks, and system tools that enable real-time monitoring, data processing, and user interaction.

The system is designed to work on Linux-based operating systems such as Ubuntu or Kali Linux. This is because the project relies on system-level networking tools like "nmcli" and "iwgetid", which are natively supported in Linux environments. While partial functionality may be adapted to other operating systems, full performance and compatibility are best achieved on Linux.

The core logic of the system is implemented using Python, which is chosen for its simplicity, readability, and strong support for system operations and web development. Python enables easy integration with system commands, file handling, and backend processing.

The application uses the Flask framework, a lightweight and flexible Python web framework. Flask is responsible for handling HTTP requests, routing, session management, and rendering dynamic web pages such as the login page, OTP verification page, and dashboard.

The user interface is developed using HTML and CSS. HTML is used to structure the web pages, while CSS is used to design and style the interface. The dashboard includes a visually appealing layout with a neon-themed design, making it easy for users to interpret network information and risk levels.

For OTP-based authentication, the system uses the SMTP protocol via Python's "smtplib" library. This allows the application to send a one-time password to the user's email for

secure login verification. A valid email account with an app-specific password is required for this feature.

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

- Login and Authentication Module
- Wi-Fi Scanning Module
- Data Processing Module
- Risk Classification Module
- Alert generation and logging module
- Connection network detection module

7.2 MODULES DESCRIPTION

The proposed Wi-Fi Intrusion Detection System is designed using a modular approach, where each module performs a specific function to ensure efficient and organized system operation. This modular design improves the clarity, maintainability, and scalability of the system. The major modules and their functionalities are described below.

The Login and Authentication Module ensures secure access to the system by validating user credentials such as username and password. It also incorporates an OTP-based verification mechanism, where a one-time password is sent to the user's email. This adds an extra layer of security and prevents unauthorized access to the system.

The Wi-Fi Scanning Module is responsible for detecting all nearby wireless networks in real time. It uses system-level commands to collect important information such as SSID,

BSSID, and signal strength. This module continuously updates the available network list to ensure that the system always has the latest data.

7.2.1 Login and Authentication Module

This module is responsible for providing secure access to the system. It verifies user credentials such as username and password. Additionally, it implements OTP (One- Time Password) verification through email to enhance security. Only authenticated users are allowed to access the dashboard and system functionalities.

7.2.2 Wifi Scanning Module

This module continuously scans nearby Wi-Fi networks using system commands. It collects essential network details such as SSID (network name), BSSID (unique hardware address), and signal strength. The scanning process is performed in real time to ensure up-to-date information about available networks.

A unique feature of this module is the implementation of honeywords. Multiple fake passwords are stored along with the real password. If a honeyword is entered, the system detects it as a malicious attempt. Instead of directly blocking the attacker, the system redirects them to a decoy dashboard, allowing monitoring of their actions.

7.2.3 Data Processing Module

The collected Wi-Fi data is processed in this module. It filters unnecessary or invalid entries, removes duplicate records, and organizes the data into a structured format. This

module ensures that only meaningful and clean data is passed to the next stage. Security measures such as session management and access restrictions are implemented to prevent unauthorized usage. This module ensures smooth and efficient management of the entire e-commerce system.

7.2.4 Risk Classification Module

This module analyzes the processed data and categorizes networks based on their risk levels. It compares detected networks with a predefined list of known networks and identifies duplicate SSIDs. Based on this analysis, networks are classified into three categories:

- Safe
- Medium Risk (Unknown Networks)
- High Risk (Fake or Duplicate Networks)

This module interacts directly with the database to perform CRUD operations efficiently. It is designed with role-based access control, ensuring that only authorized users can perform specific actions. The interface is user- friendly and organized, allowing easy navigation between different sections.

Security measures such as session management and access restrictions are implemented to prevent unauthorized usage. This module ensures smooth and efficient management of the entire e-commerce system.

7.2.5 Alert generation and Logging Module

However, this environment is completely isolated from the real system and does not provide access to actual data. All activities performed by the attacker are recorded,

including clicks, navigation patterns, and attempted actions. This information can be used to analyze attack behavior and improve security strategies.

The module plays a significant role in delaying attackers and protecting the real system from being compromised. It also helps in gathering intelligence about potential threats.

7.2.6 Connection network Detection Module

This module provides a web-based interface for users to view system outputs. It displays network details, signal strength, risk levels, and alerts in a structured and visually appealing format. The dashboard updates automatically at regular intervals to provide real-time information.

This module identifies the Wi-Fi network to which the system is currently connected. It helps users verify whether they are connected to a safe or potentially risky network.

CHAPTER 8

SYSTEM TESTING

8.1 UNIT TESTING

Unit testing focuses on testing individual modules of the system independently. Each function, such as Wi-Fi scanning, risk classification, OTP generation, and file handling, is tested separately to ensure correct operation. For example, the Wi-Fi scanning module is tested to verify whether it correctly retrieves SSID, BSSID, and signal strength. Similarly, the risk classification module is tested to ensure proper categorization of networks into safe, medium, and high risk. This testing helps in identifying and fixing errors at an early stage.

This type of testing improves code quality, reduces debugging time, and ensures that each module is reliable before combining it with others. It acts as the foundation for all other testing processes, as a system with properly tested units is less likely to fail during integration.

8.2 INTEGRATION TESTING

Integration testing is carried out to ensure that different modules of the system work together seamlessly. The interaction between modules such as scanning, data processing, classification, and dashboard display is tested. For instance, after scanning networks, the data is passed to the classification module, and then the results are displayed on the dashboard. This testing ensures smooth data flow and proper communication between modules without errors.

8.3 SYSTEM TESTING

System testing is the process of evaluating the complete and integrated system to verify that it meets all specified requirements. It is performed after all modules have been integrated and focuses on testing the system as a whole rather than individual parts. In this project, system testing ensures that the entire workflow—from login to dashboard access—operates correctly and securely.

System testing evaluates the complete application as a whole. It verifies whether the system meets all functional requirements, including login, OTP verification, real-time Wi-Fi scanning, risk classification, and dashboard updates. The system is tested under real-world conditions to ensure that it behaves as expected. This includes checking automatic page refresh, correct display of network data, and proper functioning of all features together.

8.4 SECURITY TESTING

Security testing is one of the most important aspects of this project, as it focuses on protecting the system from unauthorized access and cyber threats. It involves identifying vulnerabilities and ensuring that all security mechanisms are functioning correctly.

Security testing ensures that the system is protected against unauthorized access and vulnerabilities. The login system is tested to verify that only valid users can access the dashboard. OTP authentication is tested to ensure secure verification. Additionally, the system is checked to confirm that sensitive data, such as user credentials and session information, is handled securely, this testing ensures that the system maintainance.

Additionally, session management is tested to ensure that user sessions are secure and cannot be hijacked. The alert module is also tested to confirm that it sends notifications to the admin whenever suspicious activity is detected.

8.5 USER ACCEPTANCE TESTING

User Acceptance Testing is the final level of testing performed by end users or clients to ensure that the system meets their requirements and expectations. It focuses on usability, functionality, and overall user experience. In this project, the admin tests the system to verify that all features are easy to use and function correctly.

End users interact with the system to check its usability, interface design, and overall functionality. The dashboard is evaluated for clarity, ease of use, and responsiveness. Feedback from users is collected and necessary improvements are made to enhance user experience.

8.6 PERFORMANCE TESTING

Performance testing evaluates how efficiently the system operates under normal and expected conditions. It focuses on response time, speed, scalability, and system stability. In this project, performance testing ensures that the login process is fast and does not cause delays for users. It also checks how quickly OTPs are generated and delivered to the user's email.

Performance testing is carried out to evaluate the system's responsiveness and stability under different conditions. The system is tested for continuous Wi-Fi scanning and frequent dashboard updates. It ensures that the application performs efficiently without delays or crashes, even when multiple networks are detected. This testing confirms that the system can handle real-time operations smoothly.

CHAPTER 9

RESULT AND DISCUSSION

The results of the WiFi intrusion detection system demonstrate that the proposed model is effective in identifying unauthorized access and malicious activities within wireless networks. Using features such as packet size, signal strength, traffic patterns, and protocol behavior, the system was able to distinguish between normal and abnormal network activity with high accuracy. Experimental evaluation showed that machine learning-based approaches, particularly supervised models, achieved strong detection rates with minimal false positives.

This indicates that the model can reliably detect common attacks such as denial-of-service (DoS), spoofing, and unauthorized access attempts. Additionally, real-time monitoring capabilities improved responsiveness, allowing quicker identification of threats compared to traditional rule-based systems.

From a discussion perspective, the findings highlight the importance of feature selection and data quality in improving detection performance. While the system performed well under controlled conditions, challenges remain in handling dynamic and large-scale real-world environments where network traffic patterns constantly evolve. False alarms, though reduced, still pose a concern and may affect user trust if not properly managed. Furthermore, encrypted traffic limits visibility, making it harder to detect certain sophisticated attacks.

Future improvements could include the integration of deep learning techniques and adaptive models that continuously learn from new threats. Overall, the study confirms that intelligent WiFi intrusion detection systems play a crucial role in enhancing network security, though ongoing optimization is necessary for practical deployment.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

In this project, the WiFi intrusion detection system presented in this study proves to be an effective solution for identifying and preventing unauthorized access and malicious activities in wireless networks. By leveraging machine learning techniques and analyzing network traffic features, the system is capable of accurately distinguishing between normal and abnormal behavior. The results demonstrate high detection accuracy, reduced false positives, and the ability to detect both known and unknown threats, making it a reliable tool for enhancing network security.

Furthermore, the study highlights the advantages of using intelligent and adaptive approaches over traditional rule-based systems. The ability of the model to learn from data and adapt to new attack patterns ensures better protection in dynamic and evolving network environments. Despite some challenges such as handling encrypted traffic and maintaining performance in large-scale deployments, the overall effectiveness of the system remains significant.

Finally, this work emphasizes the importance of continuous improvement and future enhancements in WiFi intrusion detection systems. Incorporating advanced techniques such as deep learning, real-time analytics, and hybrid detection models can further strengthen system performance. With ongoing research and development, such systems can play a crucial role in safeguarding wireless communications, ensuring data privacy, and maintaining the integrity of modern network infrastructures.

10.2 FUTURE WORK

The WiFi intrusion detection systems can focus on improving detection accuracy and adaptability by incorporating advanced techniques such as deep learning and hybrid models. While current machine learning approaches perform well, deep learning algorithms like convolutional neural networks (CNNs) and recurrent neural networks (RNNs) can better capture complex patterns in network traffic. Integrating both anomaly-based and signature-based methods into a hybrid system can further enhance the ability to detect both known and unknown attacks, making the system more robust in real-world scenarios.

Another important direction for future research is the development of real-time and lightweight intrusion detection systems suitable for resource-constrained devices such as IoT-enabled WiFi networks. Optimizing models to reduce computational overhead, memory usage, and power consumption will enable deployment on edge devices without compromising performance. Additionally, implementing distributed detection mechanisms can help monitor large-scale networks more efficiently and improve response times to potential threats.

Handling encrypted traffic is also a significant challenge that needs to be addressed in future work. As more network communications become encrypted, traditional inspection methods become less effective.

Finally, future research can also focus on improving dataset quality and benchmarking standards. Creating comprehensive and realistic datasets that reflect modern WiFi network conditions will enhance model training and evaluation. Collaboration between researchers and industry professionals can lead to the development of standardized frameworks for testing and comparing intrusion detection systems. Overall, these advancements will contribute to building more secure, efficient, and scalable WiFi intrusion detection solutions.

APPENDIX - A SOURCE CODE

dashboard.html

```
<!DOCTYPE html>
<html>
<head>
<title>Wi-Fi Security Dashboard</title>
<meta http-equiv="refresh" content="5">
<link
href="https://fonts.googleapis.com/css2?family=Orbitron:wght@400;600&display=swa
p" rel="stylesheet">
<style>  body  {  background:
#0a0a0a; color:  #00ffcc; font-
family: 'Orbitron', sans-serif; text-
align: center;
}  h1  {  color:  #00ffff; text-
shadow: 0 0 15px #00ffff;
}  table { margin: auto; width:
95%; border-collapse: collapse;
box-shadow: 0 0 20px #00ffcc;
}  th,
td {
padding: 10px;
border: 1px solid #00ffcc;
}

h1 {
```

```

    color: #00ffff; text-shadow: 0
    0 15px #00ffff;
} }
tr:hover {
    background: #111;
}
.safe { color: #00ff00; }
.medium { color: #ffcc00; }
.high { color: #ff0033; }
.card { margin: 20px; padding:
    15px; border: 1px solid
    cyan; box-shadow: 0 0 15px
    cyan; border-radius: 10px;
}
</style>
</head>
<body>
<a href="/logout" style="color:red;">Logout</a>
<h1> Wi-Fi Intrusion Detection </h1>
<div class="card">
    <h3>  Connected Network</h3>
    <p>{{ connected }}</p>
</div>
<table>
<tr>
    <th>SSID</th>
    <th>BSSID</th>
    <th>Signal</th>
    <th>Strength</th>
    <th>Time</th>
    <th>Risk</th>
</tr>
{% for net in networks %}
<tr>

```

```

<td>{{ net.ssid }}</td>
<td>{{ net.bssid }}</td>
<td>{{ net.signal }}</td>
<td>{{ net.strength }}</td>
<td>{{ net.time }}</td>
<td class="
{% if 'HIGH' in net.risk %}high
{% elif 'MEDIUM' in net.risk %}medium
{% else %}safe
{% endif %}
">
{{ net.risk }}
</td>
</tr>

{% endfor %}
</table>
</body> </html>

```

dashboard.py

```

from flask import Flask, render_template, request,
redirect, url_for, session
import random
import smtplib
import subprocess
from scanner import scan_wifi, classify_risk

app = Flask(__name__)

app.secret_key = "secret123"

# Fixed login
USERNAME = "admin"
PASSWORD = "admin"

# Email config (optional)
SENDER_EMAIL = "your_email@gmail.com"
APP_PASSWORD = "your_app_password"

# ----- SEND OTP EMAIL
-----def send_otp_email(to_email, otp):
try:
    subject = "Your OTP Code"
    body = f"Your OTP is: {otp}"
    message = f"Subject: {subject}\n\n{body}"
    server = smtplib.SMTP("smtp.gmail.com", 587)
    server.starttls()

```

```

server.login(SENDER_EMAIL, APP_PASSWORD)
server.sendmail(SENDER_EMAIL, to_email,
message) server.quit() print(" OTP sent to email:",
to_email)
except Exception as e:
    print(" Email error:", e)
# ----- GET CONNECTED WIFI ----- def
get_connected_wifi(): try: result = subprocess.check_output("iwgetid -r",
shell=True).decode().strip() return result if result else "Not Connected"
except:
    return "Error"
# ----- LOGIN -----
@app.route("/", methods=["GET", "POST"]) def
login():
    print(" Login page accessed") if
    request.method == "POST":
    print(" POST request received")
    username =
    request.form.get("username")
    password =
    request.form.get("password")
    email =
    request.form.get("email")
    print("Username:", username)
    print("Password:", password)
    print("Email:", email) if
    username == USERNAME and
    password == PASSWORD:
    print(" Login successful") otp =
    str(random.randint(100000,
    999999)) session["otp"] = otp
    session["email"] = email
    # ALWAYS PRINT OTP

```

```

    print("    GENERATED  OTP:",
    otp) # Send email (optional)
    send_otp_email(email,    otp)
    return redirect("/otp")
else:
    print("        Login        failed")        return
    render_template("login.html", error="Invalid Login")
    return render_template("login.html")
# ----- OTP -----
@app.route("/otp", methods=["GET", "POST"]) def
otp():
    if request.method == "POST":
        user_otp = request.form.get("otp")
        print("Entered OTP:", user_otp)
        print("Actual OTP:", session.get("otp"))
        if user_otp == session.get("otp"):
            print("    OTP    Verified")    return
            redirect("/info") # goes to info page
        print("        OTP        Incorrect")        return
        render_template("otp.html", error="Invalid  OTP")
        return render_template("otp.html")
# ----- INFO PAGE -----
@app.route("/info")
def info():
    return render_template("info.html")
# ----- DASHBOARD -----
@app.route("/dashboard")    def
dashboard(): networks = scan_wifi()
try: with open("known_wifi.txt") as f:
    known = [line.strip() for line in f]
except:
    known = []

```

```

results    =    classify_risk(networks,
known)      connected      =
get_connected_wifi() try:
    with open("wifi_alert_log.txt") as f:
        alerts = f.readlines()
except:
    alerts = []
return render_template(
    "dashboard_neon.html",
    networks=results,
    connected=connected,
    alerts=alerts
)
# ----- LOGOUT -----
@app.route("/logout
") def  logout():
session.clear() return
redirect("/")
# ----- RUN -----
if __name__ == "__main__":
    app.run(debug=True)
knownwifi.txt HomeNetwork
OfficeWifi
MyphoneHotspot
vivo      Y27
scanner.py
import
subprocess
import  datetime
def  scan_wifi():
try:
    output = subprocess.check_output(
        "nmcli -t -f SSID,BSSID,SIGNAL dev wifi list", shell=True

```

```

).decode()
    networks = []
for line in output.split("\n"):
    if line.strip() == "":
        continue
    parts = line.split(":")
    if len(parts) >= 3:
        ssid = parts[0]
        signal = parts[-1]
        bssid = ":".join(parts[1:-1])
        if not signal.isdigit():
            continue
        networks.append({
            "ssid": ssid,
            "bssid": bssid,
            "signal": signal,
            "time": datetime.datetime.now().strftime("%H:%M:%S")
        })
# Remove duplicates
unique = {net["ssid"]: net for net in
networks}
return list(unique.values())
except Exception as e:
    print("Error:", e)
    return []
def
classify_signal(signal):
try: signal = int(signal)
except:
    return "Unknown"
if signal > 70:
    return "Strong"
elif signal > 40:
    return "Medium"
else:

```

```

    return "Weak"

def detect_duplicates(networks):
    ssid_map = {}

    for net in networks:
        ssid_map.setdefault(net["ssid"], []).append(net)
    return {k: v for k, v in ssid_map.items() if len(v) > 1}

def classify_risk(networks, known_ssids):
    results = []
    duplicates = detect_duplicates(networks)
    alerts = []
    for net in networks:
        ssid = net["ssid"]
        signal = net["signal"]
        if ssid in duplicates:
            risk = "HIGH (Fake Wi-Fi)"
            alerts.append(f'{net["time"]} - Fake WiFi detected: {ssid}')
        elif ssid not in known_ssids:
            risk = "MEDIUM (Unknown)"
            alerts.append(f'{net["time"]} - Unknown WiFi: {ssid}')
        else:
            risk = "SAFE"
        net["risk"] = risk
        net["strength"] = classify_signal(signal)
    results.append(net)  # Save
    with open("wifi_alert_log.txt", "w") as f:
        for alert in alerts:
            f.write(alert + "\n")
    return results

```

APPENDIX – B

SCREENSHOTS

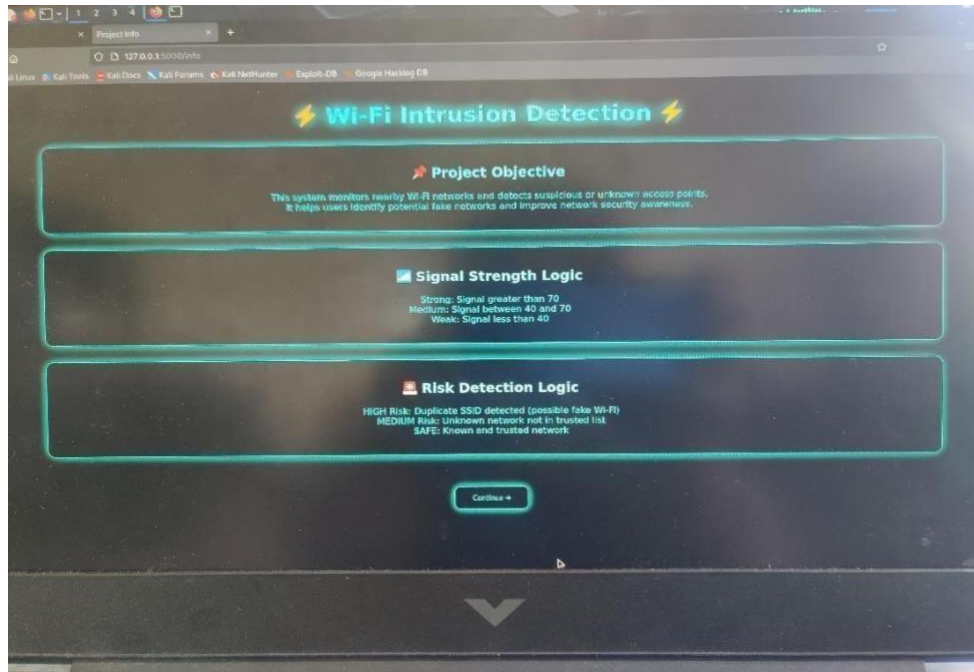


Figure B.1 Frontend

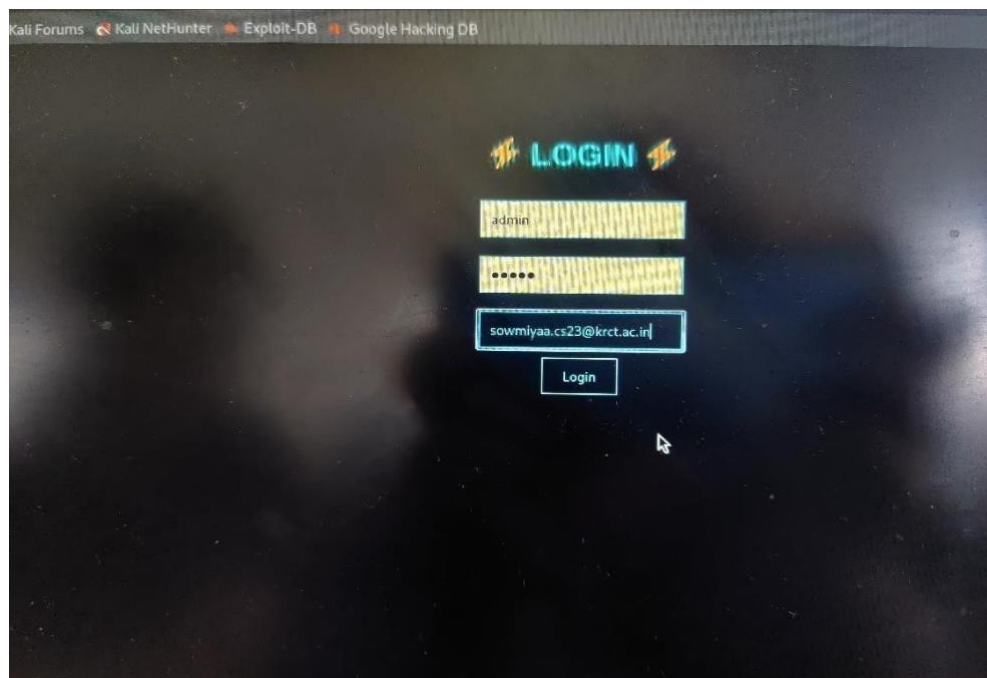


Figure B.2 Sample Image Download Window

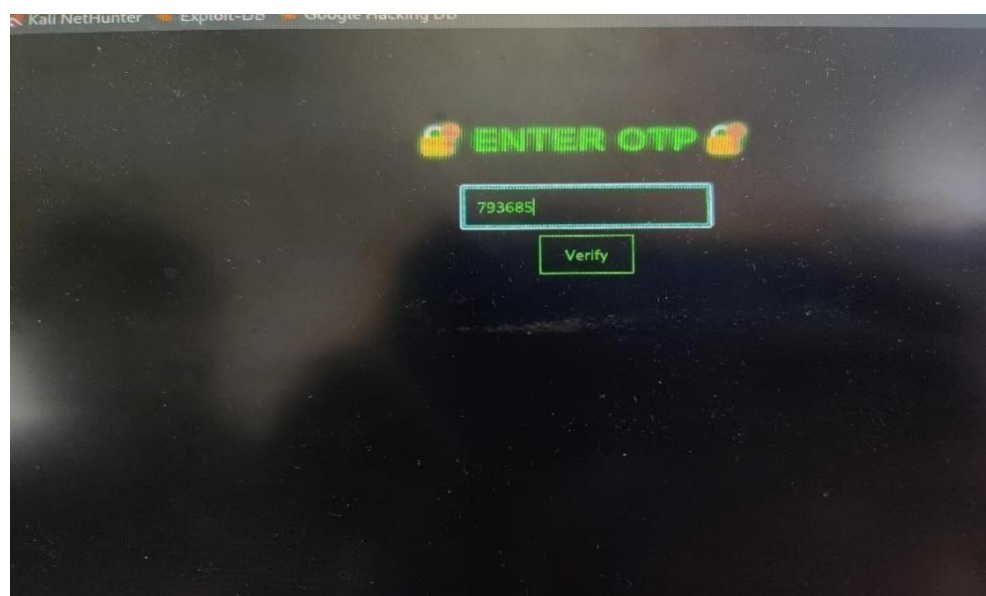


Figure B.3 Patch Classification Window

Wi-Fi Intrusion Detection

Connected Network
vivo Y21

SSID	BSSID	Signal	Strength	Time	Risk
ChiaRus TT	88:9E:84:B3:7B:98	59	Strong	09:52:11	MEDIUM ▲ (unknown)
QOO Neo6	88:9E:84:B3:7B:98	66	Strong	09:52:11	MEDIUM ▲ (unknown)
vivo Y21	88:9E:84:B3:7B:98	68	Medium	09:52:11	SAFE ■
Student	00:00:00:00:00:00	19	Weak	09:52:11	MEDIUM ▲ (unknown)
DMS	00:00:00:00:00:00	65	Medium	09:52:11	MEDIUM ▲ (unknown)
Staff	00:00:00:00:00:00	36	Weak	09:52:11	MEDIUM ▲ (unknown)
Tharu	00:00:00:00:00:00	59	Medium	09:52:11	MEDIUM ▲ (unknown)
TP-LINK_5 GHz	00:00:00:00:00:00	65	Medium	09:52:11	MEDIUM ▲ (unknown)
Harima Chan	00:00:00:00:00:00	46	Medium	09:52:11	MEDIUM ▲ (unknown)
realme 14 Pro 5G 1475	00:00:00:00:00:00	42	Medium	09:52:11	MEDIUM ▲ (unknown)
esp	00:00:00:00:00:00	42	Medium	09:52:11	MEDIUM ▲ (unknown)
Calling all Autobots	00:00:00:00:00:00	42	Medium	09:52:11	MEDIUM ▲ (unknown)

Figure 3.4 Patch Classification Results

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